

Village-Level AI Instructions: Decision Support System

Purpose

This document provides comprehensive instructions for an AI-powered (or hybrid AI + algorithmic) decision support system that analyzes village-level data to generate actionable insights for multiple stakeholders: villagers, government officials, NGOs, and researchers.

The strategy layer behind this system is to assess where households and villages are in their wellbeing journey and convert raw measurements into decision-support insights for evidence-based planning and action.

The system evaluates:

- What people currently have (assets, capabilities, access)
- What they need (gaps, vulnerabilities, constraints)
- What potentials are possible (latent capacities, regenerative opportunities)

Core Functions:

1. Calculate sustainability scores and balance sheets
2. Interpret results (identify issues, highlight successes)
3. Match resources (connect surpluses with deficits across villages)
4. Generate trajectories (create improvement roadmaps)
5. Showcase innovations (highlight best practices for replication)

CRITICAL FRAMEWORK: Asset, Liability & Flow Model

Every sub-domain uses a triple accounting structure similar to business balance sheets:

- **ASSET**: Sources, infrastructure, capacity — the sustainably renewable and reusable portion of the resource
- **LIABILITY**: Actual consumption, needs, debts, restoration costs — what is owed or consumed
- **FLOW**: The movement of resources in/out — production, consumption rates, replenishment rates, depletion rates

AI Actions When:

- **Liability > Asset**: FLAG IT → Compute resources needed to fill gap → Match with available resources/funds from surplus areas
- **Asset > Liability**: SHOWCASE IT → Match surplus with deficit areas (inter-village resource sharing)
- **Asset at risk (degraded/depleted)**: Identify restoration liabilities (degraded resources need investment to restore)
- **Flow imbalance (outflow > inflow)**: FLAG unsustainable consumption → Suggest alternatives → Generate trajectory for correction

TABLE OF CONTENTS

1. Land
2. Water
3. Forestry & Timber
4. Energy & Fuel
5. Housing
6. Community Infrastructure
7. Mobility & Transportation
8. Agriculture / Food
9. Business & Commercial Establishments
10. Income & Expenditure
11. Personal & Household Items
12. External Data Sources

1. Land

[SDGs: 11, 15 | Scope: Village Level]

Asset: Total land area + Forest cover + Untouched/pristine land

Liability: Land allocated to housing, agriculture, infrastructure + Population needs + Restoration needs (if degraded)

Flow: Rate of land use change (urbanization vs. reforestation) + Population growth impact on land demand

Data Collection & Calculation

Compute total land area belonging or demarcated as village

[Method: GIS Analysis / Village Records | Tier: **Core**]

Compute land use breakdown:

[Method: GIS Analysis + Ground Truthing | Tier: **Core**]

→ Housing

→ Agriculture

→ Roads

→ Public infrastructure

→ Commercial infrastructure

→ Forest / untouched land

Compute population density (population / total area in km²)

[Method: Census Data + GIS | Tier: **Core**]

Compute population carrying capacity based on available land at sustainable density

[Method: Algorithmic Calculation | Tier: **Extended**]

Thresholds (Bio-Region Specific)

Forest Cover: Minimum 60% of total village land

→ Source: Ecological sustainability research — forests sustain biodiversity, regulate climate, prevent soil erosion. This is a project-defined minimum; actual thresholds should be adjusted per bio-region (e.g., alpine, coastal, tropical rainforest zones may differ). Reference: International Union for Conservation of Nature (IUCN) Guidelines on Forest Landscape Restoration; Malaysian National Forestry Policy targets.

→ Bio-region variations: Coastal mangrove zones may target 40-50%, tropical inland may target 60-70%, mountainous zones may target 50-60%.

Population Density:

- Ideal sustainable range: 400–2000 per km²

→ Source: Rural settlement planning standards. Densities below 400/km² may indicate underpopulation (insufficient critical mass for services); above 5000/km² indicates overcrowding pressure on resources. Reference: UN-Habitat rural density guidelines; Malaysian Department of Town and Country Planning standards.

- Maximum threshold: 5000 per km² → Flag for immediate action
- Ideal minimum: 400 per km²

AI Directives

FLAG if:

- Forest cover < 60% (or bio-region specific threshold)
- Population density > 5000 per km²
- Insufficient land allocated for public infrastructure

SHOWCASE if:

- Forest cover > 70%
- Optimal density achieved (400–2000 per km²)
- Efficient land use (high utilization without degradation)

MATCH surplus/deficit:

- Identify villages with land surplus → Match with villages experiencing land deficit
- Suggest managed migration or resource sharing

COMPUTE:

- Resources needed for reforestation (if below forest cover threshold)
- Carrying capacity based on sustainable density
- Timeline for achieving forest cover targets

2. Water

[SDGs: 6 | Scope: Village Level + Household Level]

Asset: All water sources (rivers, groundwater, springs, rainwater, lakes, streams) $\times$ sustainable supply capacity + Storage capacity (natural + developed)

Liability: Total consumption needs (household + agricultural + industrial) + Restoration costs for polluted/degraded sources + Infrastructure expansion/maintenance costs

Flow: Sustainable supply rate (litres/year inflow) vs. Total consumption rate (litres/year outflow) + Seasonal variation + Replenishment cycles

Data Collection & Calculation

Identify all available water sources:

[Method: Field Survey + GIS | Tier: **Core**]

- River
- Groundwater
- Spring
- Rainwater
- Lake
- Streams

For each source, assess:

[Method: Field Survey + Testing | Tier: **Core**]

- Current condition (healthy / polluted / degraded)
- Whether tapped or untapped → FLAG untapped sustainable sources as potential assets
- Sustainable supply capacity (litres per year)
- Current consumption levels

Compute total water consumption needs:

[Method: Household Survey Aggregation + Estimations | Tier: **Core**]

- Drinking and cooking (per capita threshold)
- Sanitation and bathing (per capita threshold)
- Irrigation (per acre threshold — varies by crop and bio-region)
- Livestock (per animal threshold)
- Industrial/commercial uses

Compute surplus/deficit of water availability:

[Method: Algorithmic Calculation | Tier: **Core**]

- Flag if deficit exists
- **Showcase if there's a surplus** → Match with deficit villages

Assess water storage capacity:

[Method: Field Survey | Tier: **Extended**]

- Natural storage (lakes, ponds)
- Developed storage (tanks, reservoirs)
- Compute shortage/surplus
- **If shortage, compute resources and knowhow needed to fill the gap**

Assess water distribution infrastructure:

[Method: Field Audit | Tier: **Extended**]

- Piped system coverage (% of households)
- Quality of distribution network
- Losses due to leakage (if data available)
- **Whether sufficient or not → if not, compute resources needed & match with resources/funds available**
- **Showcase efficient/innovative water distribution & governance systems**

Assess wastewater and sewage management:

[Method: Field Audit | Tier: **Core**]

- Sewage treatment plant capacity
- % of houses connected to sewage system
- Discharge location and environmental impact
- Compute needs for expansion or new infrastructure **& match with available resources**
- **Showcase innovative methods if any for treatments**

Compute future water needs based on population projections:

[Method: Algorithmic Calculation | Tier: **Contextual**]

- 5-year, 10-year, 20-year, 30-year forecasts
- Required investment for infrastructure development
- **Carrying capacity — water as a limiting constraint on population**

Assess water quality at point of use:

[Method: Water Testing Kit / Lab Analysis | Tier: **Core**]

- Drinking water quality (potable / needs filtration / contaminated)
- Sanitation water quality
- Flag if quality standards not met
- **Showcase best practices if any**

Thresholds

Water Consumption: 50–100 L/person/day

→ Source: WHO Guidelines for Drinking-water Quality (4th edition, 2011). Minimum 50 L/day for basic survival needs (drinking, cooking, hygiene); 100 L/day for a dignified standard of living.

Water Quality: Potable = 1.0, Needs Filtration = 0.5, Contaminated = 0.0

→ Source: WHO water quality guidelines. Potable means meeting bacteriological and chemical parameters per WHO standards.

Scarcity Alert: >20% households reporting scarcity = medium priority, >50% = critical

→ Source: Project-defined threshold based on community resilience literature.

AI Directives

FLAG if:

- Water deficit exists (liability > asset)
- Water quality below potable standards
- >20% households report water scarcity
- Sources degraded/polluted → **Identify restoration needs** (these become restoration liabilities)

SHOWCASE if:

- Water surplus exists
- Innovative water distribution & governance systems
- Best practices for water conservation
- Effective sewage treatment methods

MATCH:

- Surplus water capacity with deficit villages
- Infrastructure needs with available resources/funds
- Innovative methods with villages needing solutions

COMPUTE:

- Resources and knowhow needed to fill water gaps
- Carrying capacity with water as limiting factor
- Future infrastructure investment needed

3. Forestry & Timber

[SDGs: 13, 15 | Scope: Village Level + Community Level]

Asset: Total forest area + Standing timber volume + Biodiversity value + Non-timber forest produce potential

Liability: Timber consumption needs + Restoration costs (for degraded/fragmented forest) + Reforestation investment + Land needed to reach 60% threshold

Flow: Annual sustainable harvest capacity (inflow/regeneration) vs. Actual annual harvest (out-flow/depletion) + Deforestation/reforestation rate

Data Collection & Calculation

Assess total forest cover within village boundaries:

[Method: GIS Analysis | Tier: **Core**]

- Forest type (natural / planted / mixed)
- Total area under forest

→ **Minimum 60% of total village land should be forest – flag if below**

Assess forest condition:

[Method: Field Survey / Ecological Assessment | Tier: **Core**]

→ Healthy

→ Disturbed

→ Degraded

→ Diseased

→ Fragmented

→ **Flag and showcase** (flag degraded conditions; showcase healthy forests)

Compute sustainable timber supply:

[Method: Forestry Inventory / Estimation | Tier: **Core**]

→ Annual sustainable harvest volume (cubic meters)

→ Actual timber harvested annually

→ Surplus / deficit

→ Flag if over-harvesting detected (outflow > regeneration)

→ **Showcase if any best practice** (e.g., Daisugi — Japanese sustainable forestry technique)

Assess timber sources:

[Method: Household Survey Aggregation | Tier: **Extended**]

→ Timber from own landholding

→ Timber from community forest

→ Timber purchased from outside

→ **Compare to sustainable harvesting threshold**

→ **Flag and showcase**

Compute timber needs for coming year:

[Method: Household Survey Estimates | Tier: **Extended**]

→ Construction purposes

→ Fuel wood

→ Furniture and crafts

→ Commercial purposes

→ **Compare with sustainable flow/capacity**

→ **Flag and showcase**

Assess other forest produce:

[Method: Community Survey | Tier: **Contextual**]

→ Types (fruits, nuts, medicinal plants, etc.)

→ Quantity harvested

→ Sustainable supply vs. actual harvesting

→ Purpose (self-consumption / sale)

→ **Flag and showcase**

Assess biodiversity and ecosystem health:

[Method: Ecological Survey | Tier: **Extended**]

- Current condition
- Threatened species (if any)
- Required investment for restoration/conservation
- Flag if significant biodiversity loss identified
- **Showcase** healthy ecosystems

Compute land to be returned to forest:

[Method: Algorithmic Calculation | Tier: **Extended**]

- Based on 60% forest cover requirement
- Identify degraded agricultural land suitable for reforestation
- Timeline for reforestation
- **Resources computation**
- **Trajectory generation and just transition** (economic/social impacts of reforestation)

Thresholds

Forest Cover: 60% minimum

→ Source: Ecological sustainability research — International Union for Conservation of Nature (IUCN) forest landscape restoration guidelines. Critical for biodiversity, carbon sequestration, watershed protection. Bio-region specific adjustments recommended (see `village_level_thresholds.md`).

Sustainable Harvest: Annual harvest $\leq$ Annual regeneration rate

→ Source: Forest management principle — sustained yield doctrine (ITTO Tropical Forest Management Guidelines). Harvest exceeding regeneration = unsustainable and depletes the asset.

Forest Health: Healthy = 1.0, Disturbed = 0.6, Degraded = 0.3, Diseased/Fragmented = 0.0–0.2

→ Source: Forest health assessment standards — FAO Global Forest Resources Assessment methodology.

AI Directives

FLAG if:

- Forest cover $< 60\%$
- Over-harvesting detected (harvest $>$ sustainable capacity)
- Forest condition degraded/diseased
- Significant biodiversity loss

SHOWCASE if:

- Forest cover $> 70\%$
- Best practices in sustainable harvesting (e.g., Daisugi)

- Healthy, biodiverse ecosystems
- Innovative forest management

MATCH:

- Timber surplus villages with deficit villages
- Reforestation resources with villages needing restoration

COMPUTE:

- Resources needed for forest restoration
- Trajectory and timeline for achieving 60% cover
- Just transition plan (if reforestation impacts existing livelihoods)

4. Energy & Fuel

[SDGs: 7, 13 | Scope: Village Level + Household Level]

Asset: Renewable energy capacity (tapped + untapped) + Grid connection infrastructure + Energy storage capacity

Liability: Total energy consumption + Grid dependency (fossil fuel portion) + Infrastructure development/maintenance costs + Carbon emissions debt

Flow: Energy generation rate (kWh/year produced) vs. Energy consumption rate (kWh/year consumed) + Fuel import dependency rate

Data Collection & Calculation

Assess grid electricity connection and consumption:

[Method: Household Survey + Utility Bills | Tier: **Core**]

- Is village connected to electric grid? (Yes/No) → **If no, then flag**
- If yes, compute yearly consumption (units/kWh) → **This is energy liability**
- Compute yearly expenditure on electricity → **This is energy financial liability**

Assess other fuel consumption:

[Method: Household Survey | Tier: **Core**]

- Petrol (litres per year, usage purposes, expenditure)
- Diesel (litres per year, usage purposes, expenditure)
- Natural gas (consumption and expenditure)
- Traditional fuels:
 - Wood (quantity, source, expenditure)
 - Biogas (if any)
 - Briquettes
 - Dung (quantity and purpose)
- **Threshold: 80% of energy should come from within 100-mile radius**
- **If not met: flag and suggest alternatives** (local renewable sources, energy efficiency measures)

Assess renewable energy capacity at village level:

[Method: Technical Assessment / Field Survey | Tier: **Extended**]

- Solar (existing capacity, potential capacity)
- Hydro (existing capacity, potential capacity)
- Wind (existing capacity, potential capacity)
- Biomass (existing capacity, potential capacity)
- Geothermal (existing capacity, potential capacity)
- **Whether tapped or untapped**
- **If surplus renewable capacity: showcase and match with deficit areas**

Compute energy self-sufficiency:

[Method: Algorithmic Calculation | Tier: **Core**]

- Total energy consumed vs. produced locally
- % dependency on central grid
- **If deficit: identify means of sufficiency and resources required**
- **Flag and showcase**

Assess energy distribution infrastructure:

[Method: Field Audit | Tier: **Extended**]

- % breakdown: Central grid fossil fuel vs. renewable
- Local renewable microgrid coverage
- Quality and reliability of supply
- **Flag and showcase**

Compute future energy needs based on population growth:

[Method: Algorithmic Calculation | Tier: **Contextual**]

- 5-year, 10-year, 20-year, 30-year projections
- Land needed for renewable capacity development
- Investment required for energy infrastructure

Assess energy efficiency and wastage:

[Method: Household Audit Sample | Tier: **Extended**]

- Are energy-efficient appliances/systems in use?
- Potential for energy savings
- Flag if excessive consumption patterns identified
- **Showcase if savings** (efficient systems, low-waste practices)

Thresholds

Grid Connectivity: 100% target (or equivalent renewable self-sufficiency)

→ Source: SDG 7 — Affordable and Clean Energy. Universal access to electricity is a UN development goal. Reference: Malaysia's National Energy Policy 2022 targets.

Renewable Energy: 50% minimum of total consumption, 80%+ = excellent

→ Source: SDG 7 + Paris Agreement climate targets. Malaysia's Renewable Energy Policy targets. 50% minimum ensures meaningful energy transition.

Local Energy: 80% from within 100-mile radius

→ Source: Project-defined threshold — localized energy systems reduce transport losses, improve resilience, and support local economy.

AI Directives

FLAG if:

- No grid connection AND no renewable energy
- <20% renewable energy use
- <80% energy from local sources
- Excessive consumption patterns

SHOWCASE if:

- High renewable energy adoption (>80%)
- Surplus renewable capacity
- Energy-efficient systems in use
- Low-carbon energy mix

MATCH:

- Surplus renewable capacity with deficit villages
- Untapped renewable potential with implementation resources

SUGGEST:

- Alternatives to fossil fuels suitable for the village
- Renewable energy options based on geography
- Energy efficiency measures

COMPUTE:

- Resources needed for renewable energy development
- Investment required for energy infrastructure

5. Housing

[SDGs: 11 | Scope: Household Level]

Asset: Total available housing units + Total built-up area + Quality of construction + Eco-friendly design value

Liability: Housing needs (population × minimum space threshold) + Amenity requirements + Maintenance/repair costs + Renovation needs for substandard housing

Flow: New construction rate vs. Deterioration rate + Population growth-driven demand + Housing turnover

Data Collection & Calculation

Compute space & privacy available per person:

[Method: Household Survey | Tier: **Core**]

- Total built-up area / number of people
- Number of rooms
- If below minimum, flag it

Assess construction materials:

[Method: Visual Inspection / Survey | Tier: **Core**]

- Walls and structure materials:
- Organic / Traditional
- Wooden
- Concrete
- Other
- Flag if not organic/eco-friendly
- **Showcase innovative eco-friendly designs and materials**

Compute surplus/deficit of available houses:

[Method: Census Data | Tier: **Core**]

- Total housing units vs. household count
- **If deficit: Flag and compute resources needed for construction**
- **If surplus: Showcase and match with villages needing housing**

Assess available housing amenities:

[Method: Household Survey | Tier: **Extended**]

- Aggregate and list the needs
- **Compute gap between current amenities and required amenities**
- **Match needs with available resources/funds**

Assess cleanliness and eco-friendliness:

[Method: Observation / Survey | Tier: **Extended**]

- Waste management systems
- Sanitation facilities
- Green building practices
- **Showcase best practices**

Assess vibratory matrix of the house:

[Method: Subjective Assessment (Scale 1-10) of design harmony, natural light, ventilation | Tier: **Contextual**]

- **Showcase high-vibration housing designs**

Thresholds

Space per Person: $25 \text{ m}^2 +$ = excellent, $15\text{--}25 \text{ m}^2$ = adequate, $<15 \text{ m}^2$ = inadequate

→ Source: UN-Habitat Adequate Housing guidelines.

Housing Surplus: $>10\%$ extra units = buffer for growth, $\pm 10\%$ = balanced, $<-10\%$ = deficit

→ Source: Housing adequacy planning standards.

Construction Materials: Organic/Traditional = 1.0, Wooden = 0.7, Concrete = 0.5, Other = 0.3

→ Source: Project-defined ecological building preference.

AI Directives

FLAG if:

- Space per person $< 15 \text{ m}^2$
- Housing deficit (units $<$ households)
- Non-eco-friendly construction materials dominant
- Inadequate amenities

SHOWCASE if:

- Space per person $> 25 \text{ m}^2$
- Housing surplus available
- Innovative eco-friendly designs
- Traditional sustainable housing methods

MATCH:

- Surplus housing with villages experiencing deficit
- Housing needs with construction resources/funds

COMPUTE:

- Resources needed to fill housing deficit
- Investment for amenity improvements
- Carrying capacity (how many people can be housed)

SUGGEST:

- Eco-friendly alternatives to conventional materials
- Space-efficient design solutions

6. Community Infrastructure

[SDGs: 3, 4, 9, 11, 16 | Scope: Village Level + Community Level]

Asset: All existing infrastructure (educational, health, civic, cultural, environmental, utilities, specialized) + Their capacity + Quality/condition

Liability: Population-based infrastructure needs + Maintenance costs + Expansion requirements + Land needed for new development

Flow: Infrastructure utilization rate (usage vs. capacity) + Deterioration/maintenance cycle +

Population-driven demand growth

Data Collection & Calculation

Assess educational infrastructure:

[Method: Facility Audit / Village Records | Tier: **Core**]

- Early Childhood Care and Development (ECCD) center: Number, capacity, land area, condition
- Primary school: Number, capacity, land area, condition
- Secondary school: Number, capacity, land area, condition
- High school: Number, capacity, land area, condition
- Skill development center: Number, capacity, land area, condition
- College/Higher education: Number, capacity, land area, condition
- Compute deficit / surplus for each level
- Flag if educational facilities inadequate for population
- **Showcase best-in-class educational infrastructure**

Assess health infrastructure:

[Method: Facility Audit / Village Records | Tier: **Core**]

- Healthcare center/clinic: Number, capacity, type of services
- Pharmacy: Availability, stock adequacy
- Emergency response capability
- Distance to nearest hospital (if outside village)
- Flag if healthcare access is inadequate
- **Match needs with available medical resources/funds**
- **Showcase innovative community health systems**

Assess civic and administrative infrastructure:

[Method: Facility Audit | Tier: **Core**]

- Town hall / Community center: Number, capacity, condition
- Post office: Availability (Yes/No)
- Bank or financial services: Availability (Yes/No)
- Police station / Security: Availability and adequacy
- Flag if critical civic infrastructure missing
- **Showcase effective governance infrastructure**

Assess cultural and recreational infrastructure:

[Method: Facility Audit | Tier: **Extended**]

- Library: Number, capacity, collection size
- Museum or cultural center: Availability
- Sports facilities: Types, capacity, condition
- Playground: Number, area, condition

- Spiritual retreats/sanctums: Number, capacity
- **Showcase vibrant cultural/recreational spaces**

Assess environmental infrastructure:

[Method: Facility Audit | Tier: **Core**]

- Sewage treatment plant: Capacity, % houses connected
- Composting facility: Capacity, coverage
- Waste segregation and recycling center: Availability, effectiveness
- Flag if waste management infrastructure inadequate
- **Showcase innovative waste management systems**

Assess utilities and services:

[Method: Facility Audit | Tier: **Core**]

- Street lights: Coverage (% of village illuminated)
- WiFi or internet access: Availability, coverage
- Water storage tanks: Number, total capacity
- Cold storage facility: Capacity, usage
- Energy/Battery storage: Capacity
- **Match utility needs with available resources**

Assess specialized infrastructure:

[Method: Facility Audit | Tier: **Extended**]

- Burial/Cremation ground: Adequacy
- Animal shelter: Availability, capacity
- Senile care center: Availability for elderly
- Disability-friendly infrastructure: Availability and adequacy
- **Showcase inclusive infrastructure designs**

For each infrastructure category, compute:

[Method: Algorithmic Calculation | Tier: **Core**]

- Current availability
- Needs based on population (population-based thresholds)
- Surplus / Deficit
- Land area required for development
- Investment required
- **Match with resources/funds available**

Assess overall infrastructure adequacy score:

[Method: Algorithmic Calculation | Tier: **Core**]

- % of essential infrastructure available
- Quality and condition of existing infrastructure
- Flag if critical infrastructure missing or in poor condition

→ **Showcase villages with comprehensive infrastructure**

Thresholds

Education: 1 ECCD per 500 pop, 1 primary per 1000, 1 secondary per 2000

→ Source: Educational access standards (UNESCO Education for All framework; Malaysian Ministry of Education).

Health: 1 clinic per 2000 pop, pharmacy within 5 km

→ Source: WHO healthcare access standards.

Sewage: 100% coverage target

→ Source: SDG 6 — Clean Water and Sanitation.

AI Directives

FLAG if:

- Any essential infrastructure missing
- Facilities inadequate for current population
- Infrastructure in poor condition
- <100% sewage coverage

SHOWCASE if:

- Comprehensive infrastructure coverage
- Best-in-class educational/health systems
- Innovative waste management
- Inclusive, disability-friendly design

MATCH:

- Infrastructure surplus with deficit villages
- Development needs with resources/funds

COMPUTE:

- Resources needed for missing infrastructure
- Investment for each category
- Land area needed for development
- Carrying capacity based on infrastructure limits

SUGGEST:

- Alternative infrastructure solutions (e.g., mobile clinics, digital libraries)
- Low-cost construction methods
- Multi-purpose facility designs

7. Mobility & Transportation

[SDGs: 9, 11, 13 | Scope: Household Level + Village Level]

Asset: Available vehicles + Road/bridge infrastructure + Public transport capacity + Parking infrastructure

Liability: Transportation needs (unmet mobility demand) + Infrastructure maintenance costs + Carbon emissions (environmental debt)

Flow: Daily/annual travel volume (km travelled) + Fuel consumption rate + Emissions generation rate + Vehicle acquisition/retirement rate

Data Collection & Calculation

Assess vehicle ownership at household level:

[Method: Household Survey | Tier: **Core**]

- Types:
- Diesel/Petrol car
- Electric car
- Cycle
- Moped
- Motorbike
- Boat
- Number of each type
- Usage patterns:
- Kms travelled within village
- Kms travelled outside village
- Purpose of travel
- Frequency of use
- **Emissions calculation from fossil fuel vehicles**

Compute vehicle needs vs. availability:

[Method: Calculation based on Demographics | Tier: **Extended**]

- Ideal number required based on population and activities
- Surplus / deficit for each vehicle type
- Required investment
- **If deficit: Flag and compute resources needed**
- **If surplus: Showcase efficient mobility systems**

Assess parking infrastructure:

[Method: Field Observation | Tier: **Extended**]

- Required parking space per vehicle type
- Total available parking space
- Compute shortage/surplus
- **Match parking needs with available land/resources**

Assess transportation connectivity:

[Method: GIS + Field Survey | Tier: **Core**]

- Road connectivity (paved / unpaved / none)
- Bridges (Number, Condition, Needed)
- Flag if connectivity is poor or limiting development
- **Compute resources needed for road/bridge development**
- **Showcase innovative low-impact road designs**

Assess public transportation access:

[Method: Village Survey | Tier: **Core**]

- Availability (buses, shared vehicles)
- Frequency and reliability
- Accessibility for elderly and disabled
- **Showcase effective public transport systems**
- **Suggest alternatives if inadequate**

Compute transportation-related emissions:

[Method: Algorithmic Calculation | Tier: **Extended**]

- Estimated carbon footprint
- Potential for reducing emissions
- **Flag if high emissions**
- **Showcase low-carbon mobility systems**

Thresholds

Road Connectivity: Paved = 1.0, Unpaved = 0.5, None = 0.0

→ Source: Infrastructure quality assessment standards (Malaysian Public Works Department).

Unmet Mobility Needs: <20% households = adequate, >50% = critical

→ Source: Project-defined adequacy threshold.

Emissions: Compare to sustainable benchmarks

→ Source: Paris Agreement per-capita emission targets; Malaysia's NDC.

AI Directives

FLAG if:

- Poor road connectivity (unpaved or none)
- >30% households reporting unmet mobility needs
- No public transportation available
- High carbon emissions from transport

SHOWCASE if:

- Efficient public transport systems
- Low-carbon mobility (electric vehicles, cycling)

- Well-maintained road infrastructure
- Accessible transport for elderly/disabled

MATCH:

- Vehicle surplus with deficit villages
- Infrastructure needs with construction resources
- Public transport models with villages needing solutions

COMPUTE:

- Resources needed for road/bridge development
- Investment for public transport systems
- Carbon emissions reduction potential

SUGGEST:

- Electric vehicle adoption
- Cycling infrastructure development
- Shared mobility solutions

8. Agriculture / Food

[SDGs: 2, 12, 15 | Scope: Household Level + Village Level]

Asset: Agricultural land + Soil health + Food production capacity + Storage capacity + Farming knowledge/skills + Irrigation infrastructure

Liability: Food consumption needs + Soil restoration costs (if degraded) + Storage deficit + Dietary gaps + Import dependency costs

Flow: Annual food production (yield per acre/year) vs. Annual food consumption (kg/person/year) + Seasonal production/consumption cycles + Food import/export rates

Data Collection & Calculation

Compute food self-sufficiency:

[Method: Production vs. Consumption Surveys | Tier: **Core**]

- Target: Fulfill all food needs at least 90% within 100-mile radius
- Calculate production vs. consumption for each food category
- Show deficit / surplus item-wise
- **If surplus: Showcase and match with deficit villages**
- **If deficit: Compute resources needed and match with available resources**

Compute efficiency of agricultural practices:

[Method: Yield Analysis / Land Use Survey | Tier: **Extended**]

- How much land would it take to achieve food self-sufficiency vs. what is being used at present
- **Flag if excess land is used** (inefficiency)
- **Showcase if less land is used** (high efficiency)
- **Suggest alternatives** for improving efficiency

Assess soil health:

[Method: Soil Testing / Visual Indicators | Tier: **Core**]

- Stable / Disturbed / Degraded / Diseased (per acre/field assessment)
- Flag if degraded → **Compute restoration resources needed**
- **Showcase regenerative agriculture practices**
- **Suggest alternatives** for soil improvement

Assess water usage for irrigation:

[Method: Usage Estimates / Flow Meters | Tier: **Extended**]

- Actual water used vs. ideal/average
- Flag if more than ideal
- **Suggest water-saving irrigation alternatives**

Assess farming practices and methods:

[Method: Household Survey | Tier: **Core**]

- Organic vs. chemical inputs
- Traditional vs. modern techniques
- Compute need for tools/machinery
- Enlist required equipment
- **Match needs with available resources/funds**
- **Showcase innovative farming methods**

Assess available storage:

[Method: Facility Audit | Tier: **Core**]

- Grain storage capacity
- Meat/Fish storage capacity
- Fruit/Vegetable storage capacity
- Compute shortage/surplus
- **If shortage: Compute resources needed for storage development**
- **Showcase innovative storage solutions**

Assess logistics access & supply chain:

[Method: Market Survey | Tier: **Contextual**]

- Compute needs if any and assess quality
- **Match supply chain needs with available services**

Compute needs as per ideal local seasonal diet:

[Method: Nutritional Analysis | Tier: **Extended**]

- Enlist items needed for nutritionally complete diet
- Consider seasonal availability and bio-region specific patterns

Compare consumption with ideal diet:

[Method: Nutritional Analysis of Consumption Data | Tier: **Extended**]

- Assess diet quality gaps
- **Flag if major nutritional gaps**
- **Suggest production shifts** to meet dietary needs

Prepare farming calendar:

[Method: Algorithmic Planning | Tier: **Extended**]

- To meet ideal dietary need
- Seasonal planting/harvesting schedule
- **Trajectory generation** for transitioning to self-sufficiency
- **Just transition planning**

Thresholds

Food Self-Sufficiency: 90%+ within 100 miles = excellent, 70–90% = good, 50–70% = adequate, <50% = poor

→ Source: Project-defined target for localized food systems. Reference: Bioregionalism literature; FAO guidelines.

Soil Health: 80%+ stable land = excellent, 60–80% = good, <60% = needs intervention

→ Source: FAO Voluntary Guidelines for Sustainable Soil Management.

Organic Farming: 80%+ = excellent, 60–80% = good, <40% = poor

→ Source: IFOAM Organics International standards.

AI Directives

FLAG if:

- Food self-sufficiency < 70%
- Soil degraded (>40% of land)
- Excess land used for current production (inefficiency)
- Major nutritional/diet quality gaps
- Storage inadequate

SHOWCASE if:

- Food self-sufficiency > 90%
- High efficiency
- Regenerative agriculture practices
- Innovative farming methods or storage

MATCH:

- Food surplus with deficit villages
- Storage capacity surplus with deficit areas
- Innovative practices with villages needing solutions

COMPUTE:

- Resources needed to achieve 90% self-sufficiency
- Soil restoration investment required
- Storage infrastructure needed
- Carrying capacity (how many people can be fed)

SUGGEST:

- Crop alternatives for better self-sufficiency
- Soil improvement methods
- Water-efficient irrigation techniques

TRAJECTORY:

- Farming calendar to meet ideal dietary needs
- Plan just transition for crop/practice shifts

9. Business & Commercial Establishments

[SDGs: 8, 9, 12 | Scope: Household Level + Village Level]

Asset: Business infrastructure + Skills/human capital + Commercial establishments + Tourism potential + Intellectual property/traditional crafts

Liability: Business operational costs + Unmet service needs + Investment requirements + Failed business losses

Flow: Annual income/revenue generated vs. Annual operational costs + Job creation/loss rate + Business formation/closure rate

Data Collection & Calculation

Identify all household businesses:

[Method: Household Survey | Tier: **Core**]

- Type of business
- Year started
- Brief description
- Location, Land/Built-up area
- Number of coworkers, Legal structure, Annual income/loss
- Sustainability and growth potential
- **Flag if business is struggling (losses)**
- **Showcase successful, sustainable businesses**

Identify business needs:

[Method: Business Survey | Tier: **Extended**]

- Skills required
- Manpower needs
- Equipment needs
- Infrastructure needs

→ Compute investment required

→ **Match needs with available resources/funds, skills, manpower**

Identify new business plans or aspirations:

[Method: Survey | Tier: **Contextual**]

→ Type of business planned

→ Feasibility assessment, Support required

→ **Match aspirations with mentorship/funding opportunities**

→ **Suggest viable business alternatives** based on local resources

Assess village-level commercial establishments:

[Method: Market Survey / Field Audit | Tier: **Core**]

→ Shops, markets, services available

→ Coverage of daily needs

→ % of needs met within village vs. requiring outside travel

→ **Showcase villages with comprehensive local commerce**

→ **Suggest missing services** that could be established

Assess tourism potential:

[Method: Resource Assessment | Tier: **Contextual**]

→ Natural attractions

→ Cultural attractions

→ Existing tourism infrastructure

→ Income from tourism

→ Potential for development

→ **Showcase successful eco-tourism/cultural tourism models**

→ **Suggest tourism development opportunities**

Thresholds

Business Activity: $\geq 30\%$ households = excellent, 20–30% = good, 10–20% = moderate, $< 10\%$ = low

→ Source: Project-defined economic diversification target. Reference: World Bank rural livelihoods diversification research.

Profitability: $\geq 80\%$ businesses profitable = excellent, 60–80% = good, $< 40\%$ = poor

→ Source: SME performance and survival rate literature.

Local Services Coverage: 80%+ daily needs met within village = excellent

→ Source: Project-defined self-sufficiency target for local commerce.

AI Directives

FLAG if:

- <10% household business activity
- <40% businesses profitable
- Critical services unavailable
- <50% daily needs met within village

SHOWCASE if:

- >30% household business activity
- >80% businesses profitable
- Comprehensive local commerce
- Successful eco-tourism or cultural tourism

MATCH:

- Business needs with available resources
- Successful business models with aspiring entrepreneurs
- Tourism potential with development resources

COMPUTE:

- Investment required for business development
- Potential income from untapped tourism
- Services needed to meet local demand

SUGGEST:

- Viable business alternatives based on local resources
- Missing services that could be established
- Tourism development opportunities

10. Income & Expenditure

[SDGs: 1, 8, 10 | Scope: Household Level]

Asset: Total household income sources + Savings + Owned assets + Income diversity

Liability: Total expenditure + Debt (repayment obligations) + Lack of emergency funds + Insurance gaps

Flow: Annual income rate vs. Annual expenditure rate + Savings accumulation/depletion rate + Debt accumulation/repayment rate

Data Collection & Calculation

Assess total household income sources:

[Method: Household Financial Survey | Tier: **Core**]

- Income from farming
- Income from employment
- Income from business
- Income from tourism

- Other income sources
- **Identify income diversity** (more sources = resilience)

Compute total annual household income:

[Method: Household Financial Survey | Tier: **Core**]

- Average, Median, Distribution (inequality)
- **Flag if high inequality**
- **Showcase villages with equitable income distribution**

Assess total household expenditure:

[Method: Household Financial Survey | Tier: **Core**]

- Food
- Clothing
- Personal care
- Education
- Healthcare
- Travel
- Utilities
- Debt repayment
- Other expenses

Compute household financial balance:

[Method: Algorithmic Calculation | Tier: **Core**]

- Surplus / deficit
- Average savings per household
- % of households with surplus vs. deficit
- Flag if significant % in deficit (>40%)
- **Showcase villages with >80% surplus households**

Assess financial security and resilience:

[Method: Household Survey | Tier: **Extended**]

- Emergency fund availability (months of expenses)
- Debt burden (manageable / heavy / overwhelming)
- Access to credit/financial services
- Financial literacy levels
- **Flag if >50% households lack emergency funds**
- **Flag if debt burden is overwhelming for >30% households**
- **Showcase financial literacy initiatives**

Identify investment needs and opportunities:

[Method: Household Survey | Tier: **Contextual**]

- Infrastructure needs

- Business needs
- Education needs
- Housing needs
- **Match investment needs with available funds/programs**

Compute village-level economic indicators:

[Method: Aggregation | Tier: **Core**]

- Average per capita income
- Total village economic output
- Economic diversity (number of income sources)
- Self-sufficiency ratio
- **Showcase villages with high local income generation**
- **Suggest income diversification opportunities**

Thresholds

Household Surplus: $\geq 80\%$ households in surplus = excellent, 60–80% = good, 40–60% = moderate, $< 40\%$ = poor

- Source: Project-defined financial health benchmark.

Emergency Fund: 3–6 months of expenses = adequate

- Source: Financial resilience literature.

AI Directives

FLAG if:

- $> 40\%$ households in deficit
- $> 50\%$ households lack emergency funds
- High income inequality
- $> 30\%$ households with overwhelming debt

SHOWCASE if:

- $> 80\%$ households in surplus
- Equitable income distribution
- High financial literacy
- Diverse income sources

MATCH:

- Investment needs with available funds/programs
- Income-generating opportunities with unemployed/underemployed individuals

COMPUTE:

- Resources needed to bring deficit households to surplus
- Investment required for income diversification

SUGGEST:

- Alternative income opportunities
- Debt management strategies
- Financial literacy education programs

11. Personal & Household Items

[SDGs: 3, 12 | Scope: Household Level]

Asset: Local production capacity + Traditional knowledge/skills + Production infrastructure + Raw material availability

Liability: Consumption needs (personal care, cleaning, medicine, kitchen items) + External dependency costs + Lost traditional knowledge

Flow: Annual consumption rate vs. Local production rate + Import dependency rate + Traditional knowledge transfer rate

Data Collection & Calculation

Assess yearly consumption of personal care & cleaning items:

[Method: Household Survey | Tier: **Extended**]

- Soap, shampoo, toothpaste, etc.
- Detergents, disinfectants, etc.
- Quantity consumed
- % locally produced vs. purchased
- **Showcase if high % locally produced (>80%)**

Assess medicine consumption:

[Method: Household Survey / Health Records | Tier: **Core**]

- Common medicines needed
- Availability within village (pharmacy/clinic)
- % of medical needs met locally
- Flag if critical medicines unavailable locally
- **Match medicine needs with local production or supply opportunities**

Assess kitchen items and consumables:

[Method: Household Survey | Tier: **Extended**]

- Utensils, cookware replacement needs
- Locally produced vs. purchased
- **Showcase traditional kitchenware production**

Identify items that can be locally prepared:

[Method: Survey of Knowledge Holders | Tier: **Contextual**]

- Traditional products (soaps, oils, balms)
- Potential for local production to reduce dependency

- **Showcase traditional production knowledge**
- **Suggest local production opportunities**
- **Match production capacity with consumption needs**

Thresholds

Local Production: 80%+ of 6 categories = excellent, 60–80% = good, 40–60% = moderate, <40% = high external dependency

- Source: Project-defined self-sufficiency target.

AI Directives

FLAG if:

- <40% local production (high external dependency)
- Critical medicines unavailable locally
- Lost traditional production knowledge

SHOWCASE if:

- >80% local production
- Traditional production methods preserved
- Local production surplus

MATCH:

- Local production surplus with deficit villages
- Traditional knowledge with villages wanting to establish production
- Production capacity with consumption needs

COMPUTE:

- Resources needed to establish local production
- Potential cost savings from local production

SUGGEST:

- Items that can be locally produced
- Traditional production methods to revive
- Alternatives to imported products

12. External Data Sources

The AI model should integrate structured external data to validate, supplement, benchmark, and operationalize the village-level data.

Use `thresholds/village_level_thresholds.md` as the canonical benchmark and threshold layer inside this workspace. External sources should help localize, validate, or gap-fill those benchmarks rather than override them silently.

Structured Data Source Categories

- **Survey-based primary data:** Household and village responses captured in the app and field tools.
- **Community-validated participatory data:** Village-level review, moderator validation, and locally confirmed records.
- **Government registries and administrative datasets:** Household, land, occupation, education, health, and infrastructure systems.
- **Standardized global registries and code lists:** Shared taxonomies and dropdown standards for interoperability.
- **Remote sensing and GIS layers:** Satellite imagery, land cover, terrain, hydrology, and hazard maps.
- **IoT and field equipment:** Water meters, soil sensors, air-quality monitors, climate stations, and test kits.
- **Threshold and benchmark references:** WHO, UN-Habitat, FAO, IUCN, SDG frameworks, World Bank benchmarks, and Malaysian standards already reflected in `village_level_thresholds.md`.

Primary External Data

Data that directly affects scoring, validation, or threshold localization.

Domain	Source	Data Type	Usage
Land / Geo	JUPEM (Dept of Survey and Mapping Malaysia)	Geospatial boundaries, Topography	Verify area, determine slope suitability
Land / Geo	Department of Forestry	Official Forest Reserves maps	Cross-verify forest cover
Demographics	DOSM (Dept of Statistics Malaysia)	Census data (Population, Age, Gender)	Validate village population counts
Economics	DOSM	Income benchmarks (B40/M40/T20), Poverty Line Income (PLI)	Benchmark household income and surplus thresholds
Health	Ministry of Health (MOH)	Disease prevalence, Facility locations	Validate health access data, identify endemic risks
Water / Climate	MetMalaysia	Rainfall patterns, Flood risk zones	Assess water asset potential, flood vulnerability
Water / Basin	DID / JPS	Watershed, irrigation, and flood-control datasets	Validate seasonal water stress and irrigation potential
Planning / Infrastructure	PLANMalaysia / Local Authorities	Settlement planning and infrastructure norms	Validate density, facility access, and land-use assumptions

Secondary External Data

Data for context, benchmarking, and standards.

Organization	Purpose
WHO (World Health Organization)	Water quality, sanitation, and health-access standards
FAO (Food and Agriculture Organization)	Agricultural best practices, dietary diversity standards, soil-health guidelines
UN-Habitat	Housing and settlement planning standards
IUCN (International Union for Conservation of Nature)	Nature-based solutions, restoration, and biodiversity references
World Bank	Economic indicators and financial-inclusion benchmarks
Malaysia Government Sectoral Standards	Country-specific planning, forestry, agriculture, and service-access standards

Benchmark Linkage to `village_level_thresholds.md`

Use the threshold file as the authoritative internal benchmark map for:

- **Land:** UN-Habitat settlement guidance, ecological forest-cover references, and Malaysian planning norms
- **Water:** WHO water quantity and quality standards, plus Malaysian climate and basin context
- **Housing:** UN-Habitat adequate housing and space-per-person references
- **Agriculture / Food:** FAO and ag-sustainability references for food, soil, and diversity metrics
- **Business / Income:** World Bank, Malaysia SME, and local economic benchmark references
- **Ecology / Forestry:** IUCN, forestry policy, and restoration references

Integration Rules

1. **Priority:** Collected ground data > External data (unless ground data is flagged as unreliable).
2. **Validation:** Use external data to flag outliers (e.g., if village reports 100% forest cover but Satellite says 20%).
3. **Gap Filling:** If "Contextual" or "Extended" data is missing, use regional averages from external sources as proxies (clearly labeled as *Estimated*).
4. **Traceability:** Every externally sourced value should record source name, access date, geography, and whether it is measured, inferred, or estimated.

Potential API / Data-Link Integrations

Where machine-readable access exists, explore integration pathways for:

- **Government registries:** household, land, occupation, education, health, and infrastructure systems
- **Remote sensing feeds:** satellite imagery, land cover, flood-risk, and terrain services
- **Financial inclusion systems:** account access, credit history, or formal-finance proxies where lawful and consented

- **Market linkage platforms:** commodity prices, buyer networks, logistics, and local market access
- **Knowledge hubs:** agriculture extension, health guidance, vocational training, and best-practice repositories

QUICK REFERENCE: SUMMARY MATRIX

#	SUB-DOMAIN	SDGs	SCOPE	KEY THRESHOLD	THRESHOLD SOURCE
1	Land	11, 15	Village	60% forest, 400–2000/km ²	IUCN, UN-Habitat
2	Water	6	Village + Household	50–100 L/person/day	WHO
3	Forestry & Timber	13, 15	Village + Community	60% cover, harvest ≤ regeneration	IUCN, ITTO, FAO
4	Energy & Fuel	7, 13	Village + Household	50% renewable min, 80% local	SDG 7, Paris Agreement
5	Housing	11	Household	15–25 m ² /person	UN-Habitat
6	Community Infra	3, 4, 9, 11, 16	Village + Community	Population-based ratios	UNESCO, WHO, SDG 6
7	Mobility	9, 11, 13	Household + Village	<20% unmet needs	Project-defined, Malaysia NDC
8	Agriculture / Food	2, 12, 15	Household + Village	90% within 100 miles	FAO, Bioregionalism
9	Business	8, 9, 12	Household + Village	30% activity, 80% profitable	World Bank, Malaysia SME Corp
10	Income & Expenditure	1, 8, 10	Household	80% surplus households	World Bank, Malaysia DoS
11	Personal Items	3, 12	Household	80% local production	Project-defined

Last Updated: February 13, 2026 | By Yagyaansh

Threshold References: See detailed sources within each sub-domain section and in `thresholds/village_level_thresholds.md`